



Document Title	Data Analyst Team Report
Project	Cement Demand Forecasting Across Multiple Sites
Prepared For	Project Manager
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1. Project Objective

Midlands Infrastructure Group (MIG) operates **30 construction sites** across multiple regions and has been facing challenges such as inconsistent cement consumption, stockouts, overcapacity, and unpredictable demand patterns. These issues made it difficult for the business to plan deliveries efficiently and prepare for accurate forecasting.

Using the dataset provided, the Data Analyst team was responsible for examining MIG's current operational performance and uncovering the patterns behind these challenges. Our goal was to transform the raw data into clear, actionable insights and build interactive dashboards that show how each site is performing today. This work provides MIG with a strong analytical foundation, helping the business understand its current state and positioning it to successfully integrate and interpret future forecasting results.

2. Summary of Weeks 1 & 2

During Week One, the team focused on understanding the business problem, reviewing the case study brief, and exploring the dataset. We cleaned and structured the data to ensure accuracy and consistency, identified the key operational metrics, and mapped out how each metric contributes to forecasting and performance monitoring. This foundation ensured that the dashboards built in Week Two would be reliable and ready to receive the forecasting output.

In Week Two, the team developed two interactive dashboards: one for executives and one for site managers. These dashboards were designed to help MIG understand consumption trends, operational risks, and customer behaviour patterns across regions. We added slicers, bookmarks, and page navigation to make the dashboards easy to explore, and we implemented Row-Level Security (RLS) so each site manager sees only the data relevant to their site.

3. Executive Dashboard

The Executive Dashboard provides a high-level view of performance across all 30 sites and regions. It highlights total planned pours, deliveries, and consumption, helping leadership monitor efficiency and identify gaps. Monthly consumption trends reveal seasonal fluctuations, while cement type distribution shows balanced usage across CEM I, II, and III. The dashboard also identifies the top sites with pour disruption and overcapacity, enabling executives to quickly spot operational bottlenecks and prioritize interventions. Overall, it supports strategic decision-making and prepares leadership to interpret and apply forecasting outputs effectively.

Key Insights

- High-level visibility across all sites and regions
- Seasonal consumption fluctuations
- Balanced cement type usage
- Sites with highest disruption and overcapacity
- Clear indicators of operational inefficiencies



4. Site Manager Dashboard

The Site Manager Dashboard provides a localized view of each site's performance, allowing managers to understand their operational behaviour in detail. It compares planned versus actual pours, deliveries versus consumption, and silo capacity utilization. Regional behaviour analysis shows clear differences across East, South, West, and North, helping managers understand how their site compares to others. Row-Level Security ensures each manager sees only their own site, promoting accountability and focused decision-making. This dashboard equips site managers with the insights needed to reduce waste, improve planning, and align their operations with the forecasting model.

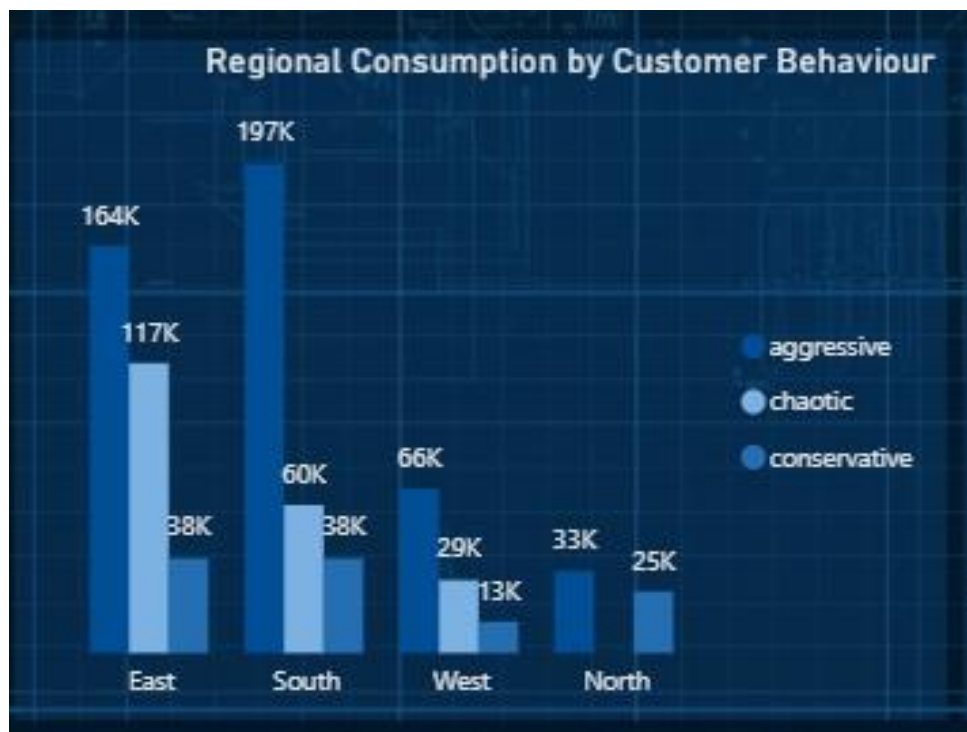
Key Insights

- Site-level performance and behaviour patterns
- Planned vs actual pours and consumption alignment
- Silo utilization and storage efficiency
- Regional behaviour differences (chaotic, aggressive, conservative)
- Secure, site-specific access through RLS



5. Final Insights

The dashboards revealed clear behavioural and operational patterns across regions. The East region recorded the highest total consumption and the highest level of chaotic behaviour, making it the most operationally demanding. The South region showed the highest aggressive consumption and the second-highest chaotic behaviour, indicating high volatility. The West region displayed moderate consumption with balanced behaviour, while the North region showed no chaotic behaviour at all, making it the most stable and predictable region.



Operationally, deliveries exceeded consumption by a significant margin, suggesting over-delivery and potential waste. High pour disruption and overcapacity days

highlighted scheduling and silo-management inefficiencies, while low silo utilization indicated underuse of available capacity.

6. Recommendations

To enhance operational efficiency and prepare for the forecasting model, the team recommends the following actions:

1. Strengthen forecasting efforts in regions that are chaotic and aggressive.
2. Align deliveries more closely with actual consumption patterns.
3. Prioritize logistics support for high-risk sites.
4. Provide training for site managers to help them interpret behavioral patterns and adjust ordering schedules as needed.
5. Use dashboards regularly to monitor performance trends and support decision-making.

The work completed by the team ensures that MIG now has a clear, data-driven view of cement demand and operational performance across all sites. The dashboards provide the structure needed to integrate the forecasting model, making it easier for the client to understand future demand and plan deliveries more effectively. This positions MIG to reduce waste, prevent stockouts, and improve operational efficiency across the entire network.